

Technical Report: Speed Dating Experiment

1. Introduction

Research Question: To what extent do individuals' pre-date stated partner preferences before a speed-dating event predict whether they form mutual matches?

This project explores what factors predict whether two people form a mutual match in a speed dating context. Using data from a large-scale speed dating experiment, we examine whether demographic characteristics, stated dating preferences, and post-date impressions can explain match outcomes.

Specifically, we compare three types of models:

- (1) models based only on demographic information,
- (2) models based on participants' self-reported dating preferences and theories, and
- (3) models based on ratings given after an actual interaction.

This analysis helps distinguish between what people say they value in a partner and what actually influences their decisions in real social interactions. Understanding these patterns can provide insight into human decision-making, metacognition, and how stated beliefs align (or fail to align) with behavior.

2. The Data Set

This project uses data from a large speed-dating study in which participants attended organized dating events and recorded their preferences, impressions, and outcomes for each interaction. The dataset contains 8,377 individual dating interactions collected across multiple event waves, with each row representing a single date between two participants. The primary outcome of interest is whether a mutual match occurred, defined as both participants indicating interest in seeing each other again.

Available Variables

- Outcome variable
 - Match: 1 = both said yes, 0 = otherwise
- Demographic information
 - Participant and partner age (age, age_o)
 - Participant and partner race (race, race_o)
 - Gender

- Pre-date preference weights (dating theories)
 - Participants' self-reported importance weights (out of 100 points) for key partner attributes before the event:
 - Attractiveness, sincerity, intelligence, fun, ambition, shared interests (attr1_1, sinc1_1, intel1_1, fun1_1, amb1_1, shar1_1)
- Partners' reported importance weights for the same attributes: (pf_o_att, pf_o_sin, pf_o_int, pf_o_fun, pf_o_amb, pf_o_sha)
- Post-date scorecard ratings
 - Ratings given after each interaction capturing participants' actual impressions of their partners:
 - Attractiveness, sincerity, intelligence, fun, ambition, shared interests (attr, sinc, intel, fun, amb, shar)
 - Corresponding partner-rated versions: (attr_o, sinc_o, intel_o, fun_o, amb_o, shar_o)

Pre-processing Steps

Several preprocessing steps were required before analysis:

- The dataset required specifying Latin-1 encoding to correctly load non-UTF8 characters.
- An initial descriptive row was skipped when reading the data.
- Analyses were restricted to waves 1–5 to ensure consistency across events.
- Identifier variables (participant IDs and wave indicators) were retained for tracking but not used as predictors.
- Due to substantial missingness in some variables, modeling was conducted on a filtered subset of the data:
- Only participants with complete data for the variables used in each model were included.
- For modeling consistency, analyses focused on one gender group to avoid duplicates or unnecessary information.

Data Quality Considerations

There are several important data quality considerations:

- There were much less matches than non-matches, so there was a large class imbalance which affects accuracy-based metrics and we needed to be more careful interpreting the model performance.
- Some variables contain high levels of missing data, limiting the usable sample size for modeling.

- Because participants appear in multiple dates, observations are not fully independent, which may inflate apparent performance for simpler models.
- The dataset contains very few Black participants relative to other racial groups, so they were underrepresented so race becomes a weak and unstable predictor
 - Limits what we can conclude about race and attraction

Despite these limitations, the dataset provides a rich opportunity to compare what people say they value, what they experience during interactions, and what ultimately predicts mutual romantic interest.

3. Results

Model A: Demographics-Only Decision Tree

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Model A - Decision Tree
Accuracy: 0.7987421383647799
ROC AUC: 0.562007874015748

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	precision	recall	f1-score	support
0	0.80	1.00	0.89	127
1	0.00	0.00	0.00	32
accuracy			0.80	159
macro avg	0.40	0.50	0.44	159
weighted avg	0.64	0.80	0.71	159

- Accuracy $\approx$ 0.80
- ROC AUC $\approx$ 0.56
- Precision and recall for the match class were 0.00

The relatively high accuracy is misleading and driven by class imbalance (most speed dates did not result in a match). The decision tree overwhelmingly predicted the majority class (no match), failing to correctly identify matches. The ROC AUC close to 0.6 indicates performance barely above chance. Precision and recall being zero for matches confirms that demographics alone provide almost no predictive signal for mutual attraction.

So this showed that demographic variables do not form meaningful decision rules for predicting matches. This model serves mostly as a baseline, but it does show that surface-level traits are insufficient for explaining romantic outcomes.

Model B: Pre-Date Preference Weights

*Although both analyses are labeled “Model B,” they have different modeling approaches and thus result in different performance metrics. In the Speed Dating Experiment notebook, Model B uses regularized logistic regression, which models match probability as a linear combination of pre-date preference weights. In contrast, the Decision Trees notebook implements a nonlinear decision tree classifier that identifies hierarchical decision rules.

Model B examined whether participants’ stated dating theories (self-reported importance weights for attributes such as attractiveness, fun, intelligence, sincerity, ambition, and shared interests) could predict mutual match outcomes. When implemented using logistic regression with regularization, the model achieved moderate performance (Accuracy ≈ 0.65 , ROC AUC ≈ 0.69), indicating that pre-date preference weights contain some predictive signal. The ROC AUC above chance suggests the model was able to meaningfully rank match versus non-match outcomes, even though overall classification accuracy remained limited.

In contrast, a decision tree version of Model B produced higher accuracy but lower ROC AUC, largely due to predicting the majority “no match” class and failing to reliably identify matches. This discrepancy highlights that while pre-date preferences may contribute small, distributed effects across many variables, they do not form strong, rule-based decision pathways. Overall, these results suggest that participants’ stated dating theories weakly align with actual match behavior, supporting the idea that people don’t actually know the factors that ultimately drive their romantic decisions (or even if they think they do, their decisions say otherwise).

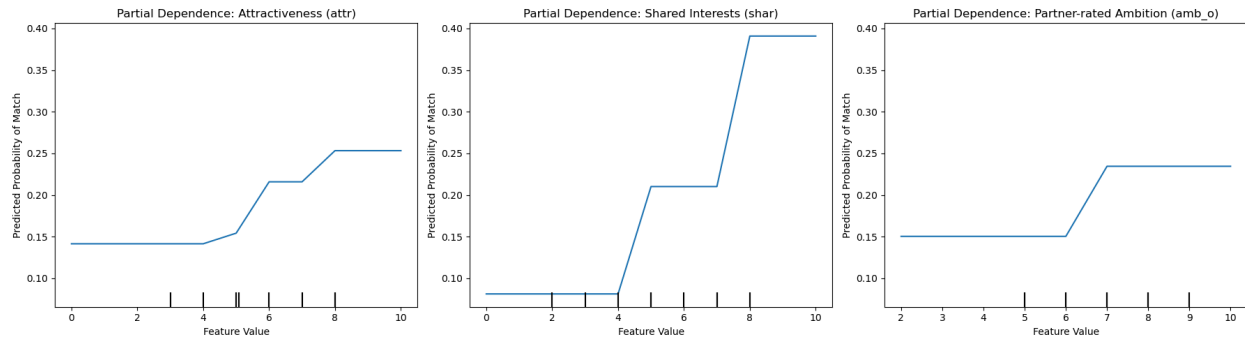
Model C: Post-Date Ratings (Decision Tree)

Performance:

- Highest predictive performance among all models
- Improved ROC AUC relative to Models A and B
- Decision tree splits were interpretable and aligned with intuitive dating dynamics

Permutation importance showed that attractiveness, shared interests, and partner-rated ambition were the most influential predictors. The partial dependence plots revealed clear increases in match probability as these ratings increased.

Interestingly, what people experience during the interaction matters far more than what they think they want beforehand. Post-date perceptions capture real-time chemistry and social evaluation, which are central to romantic decision-making.



Lasso:

- All variables zeroed out, so with lasso, no single preference was strong enough to survive regularization

4. Discussion

Across models, our analyses suggest that what people say they value before a date is only weakly related to whether a mutual match actually occurs, while post-date impressions are substantially more informative. Demographic variables alone (Model A) performed poorly, largely predicting non-matches due to class imbalance and offering little insight into romantic decision-making. This indicates that surface-level characteristics such as age and race are insufficient for explaining mutual attraction in this context.

Model B, which incorporated pre-date preference weights (participants' stated dating theories and their partners' stated preferences), showed modest improvements over demographics alone but remained limited in predictive power. While certain features such as fun and intelligence emerged as slightly more informative, overall performance suggests a gap between individuals' theories about what they value and their actual romantic choices. This misalignment highlights the role of imperfect self-knowledge and metacognition in social decision-making.

Model C demonstrated the strongest performance, with higher ROC AUC and more meaningful decision boundaries. Analyses using permutation importance and partial dependence plots revealed that post-date perceptions of attractiveness, shared interests, and partner-rated ambition were the most influential predictors of a match. These results suggest that lived experience during the interaction, rather than abstract preferences formed beforehand, better explains mutual attraction.

Taken together, these findings imply that romantic decision-making is context-dependent and experiential, rather than driven primarily by stable, pre-declared preferences. For a lab or applied setting (such as matchmaking platforms), this suggests that systems relying heavily on self-reported preferences may be less effective than those incorporating behavioral feedback or post-interaction evaluations. While the models are limited by class imbalance and sample composition, the consistent pattern across methods increases confidence in the conclusion that how people feel after interacting matters more than what they think they want beforehand.